

Understanding Agentic AI: Concepts, Architecture, and Risks

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Artificial Intelligence

The science of training machines to perform tasks that typically require human intelligence

Machine Learning

A subset of artificial intelligence utilizing algorithms that learn from examples and experiences

Deep Learning

A subset of machine learning where the learning phase is done through a neural network

Generative AI

A subset of deep learning focused on creating new content

Large Language Model (LLM)

Type of deep learning model trained on massive amounts of data

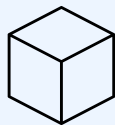
Technical Evolution

From
Standalone
Models

to

Agentic
Systems

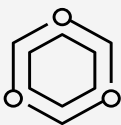
AI that can
create for you



Models

- Problem-solving
- Logical thinking
- Pattern matching

AI that can
chat for you



Assistants

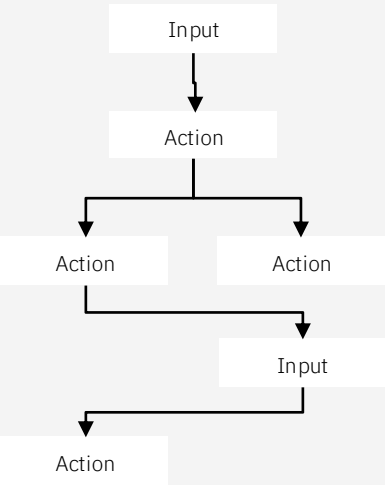
- Information retrieval
- Prescriptive tasks
- Single-step processes

AI that can
do for you

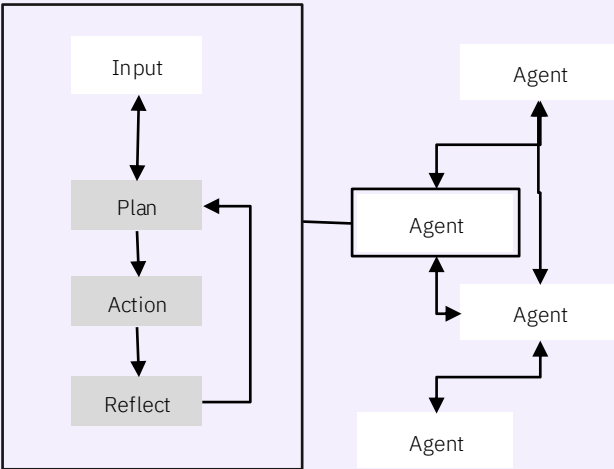


Agents

- Multi-step processes
- Autonomous actions
- Self-corrections



Traditional assistants



Single-agent assistants

Multi-agent assistants

Agentic AI - the next generation of AI

AI proficient in a domain that can act as part of a virtual workforce

In Agentic AI:

- Agents use tools and collaborate with other agents.
- Plan and act on tasks in response to a question or prompt.
- Reflect on the results of its actions.
- Learns iteratively, refining its approach to better align with its defined objectives.

Can you help me book a flight from Albuquerque to New York City for May 25?

OK, here are the best flight options from your closest airport:

New York (JFK)

1 stop \$320
4 h 40 min

New York (LGA)

1 stop \$370
4 h 32 min

Would you like me to book your flight?

Expert Agents

AI



Flight scheduler agent

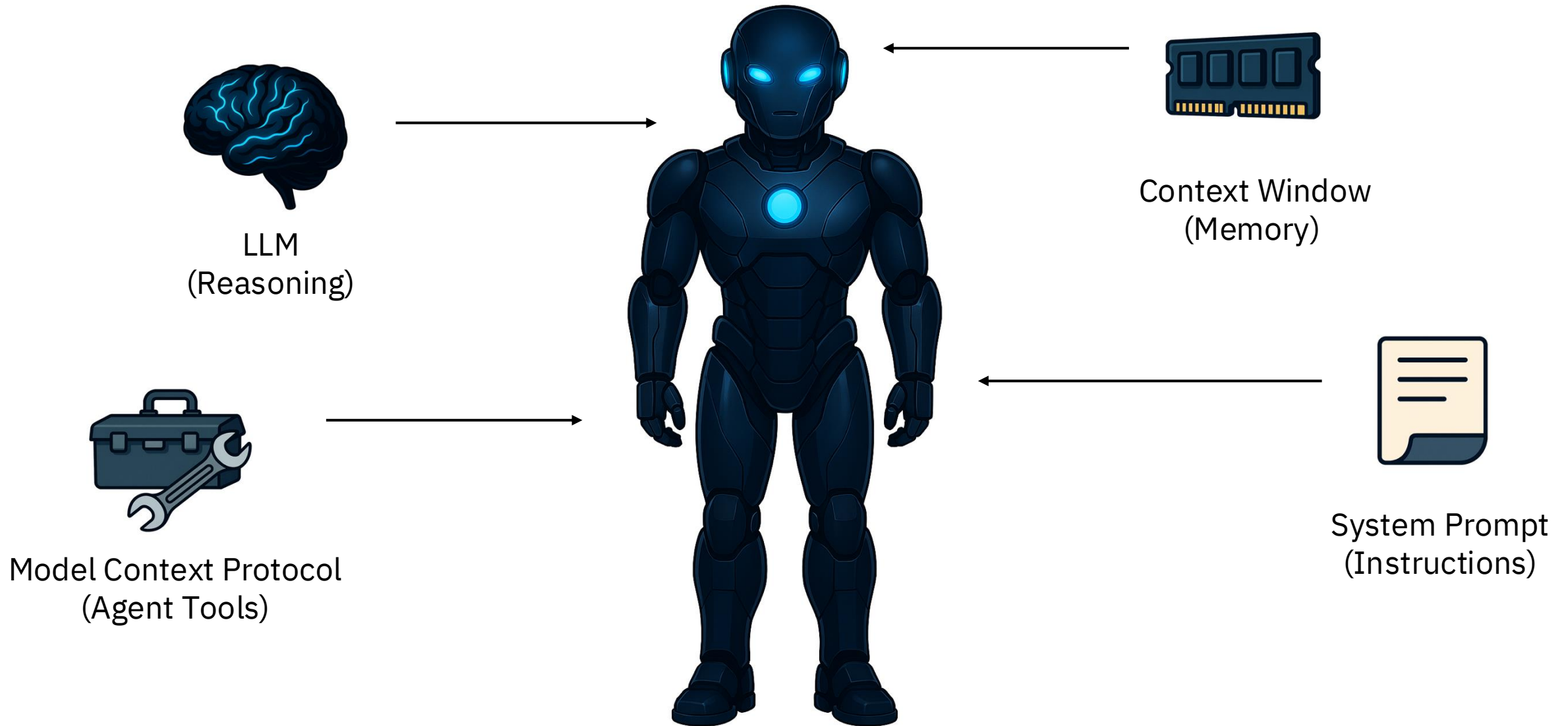


Hotel booking agent



Local activities agent

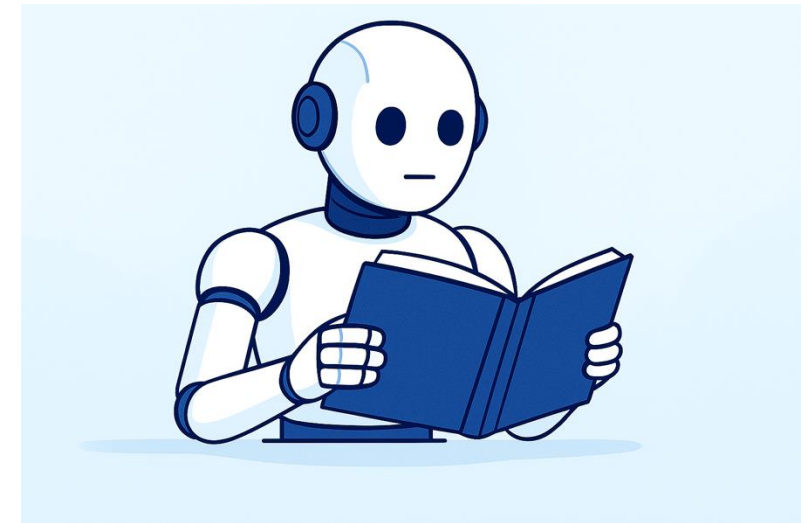
Anatomy of an AI Agent



What an LLM “knows”

Parameters

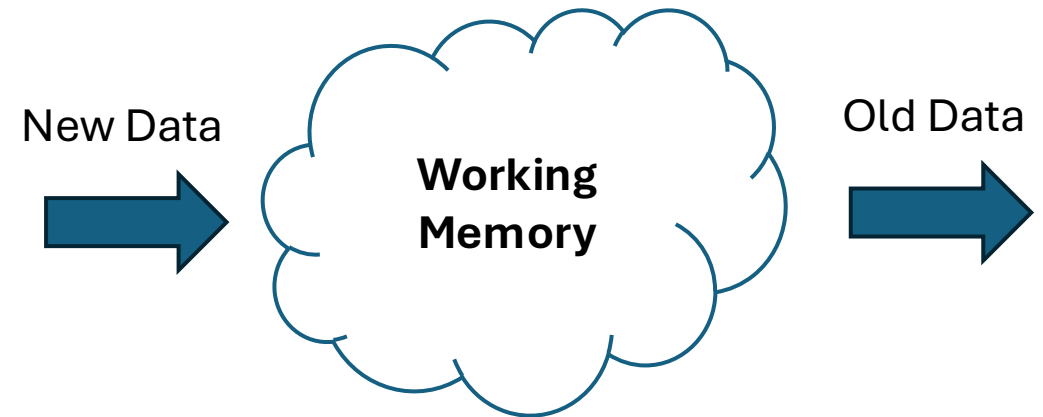
- **Learned numbers:** Numerical weights and biases learned during training
- **Encode knowledge:** Capture language patterns, relationships and reasoning behaviors
- **Fixed at runtime:** Parameters do not change when the model runs
- **Drive predictions:** Together with prompt and context, parameters determine the model’s next token output



What an LLM “knows”

Context Window

- **Active Working Memory:** The context window is limited amount of text an LLM can see and reason over at one time.
- **Includes the conversation:** Contains the prompt, prior instructions, system instructions, and any tool outputs.
- **Determines awareness:** Everything the model “knows” during a response must be inside this window
- **Finite and shifting:** When the window fills up, older information falls out and is no longer accessible.



Retrieval Augmented Generation (RAG)

- Combines LLM generation with real-time retrieval of external knowledge
- Grounds model responses in trusted, up-to-date data sources
- Reduces hallucinations and improves accuracy and explainability
- Enables agents to reason, act, and respond using enterprise or domain data



Why Transformers Matter

Attention enables reasoning

- Models focus on what matters across the entire context
- Links instructions, goals, and information dynamically
- Enables multi-step reasoning

Scaling enables planning

- Larger Models exhibit chain-of-thought behavior
- Complex decision patterns emerge with scale
- Planning is *learned*, not programmed



Without transformers, we would have chatbots - not agents

Tokenization

Breaks text into smaller parts for easier machine analysis, helping machines understand human language.

Types:

- Word tokenization
- Character tokenization
- Subword tokenization
- Sentence tokenization
- N-gram tokenization
- ..



IBM Has a Roadmap to a 'Fault-Tolerant'
Quantum Computer by 2029.

20 Tokens 66 Characters

Categorization / Named Entity Recognition

“Organization” entity type

IBM Has a Roadmap to a 'Fault-Tolerant'
Quantum Computer by 2029.

“Date” entity type

System Prompt

The **system prompt** is a set of instructions given to an AI agent that defines:

Identity & Role

- Communication style

Capability and tools

- How to use each tools, examples, and expected outputs

Behavior Guidelines

- How to format responses
- Error handling strategies
- When to use certain tools

Domain knowledge

- Technical concepts
- Best practices

System Prompt Example

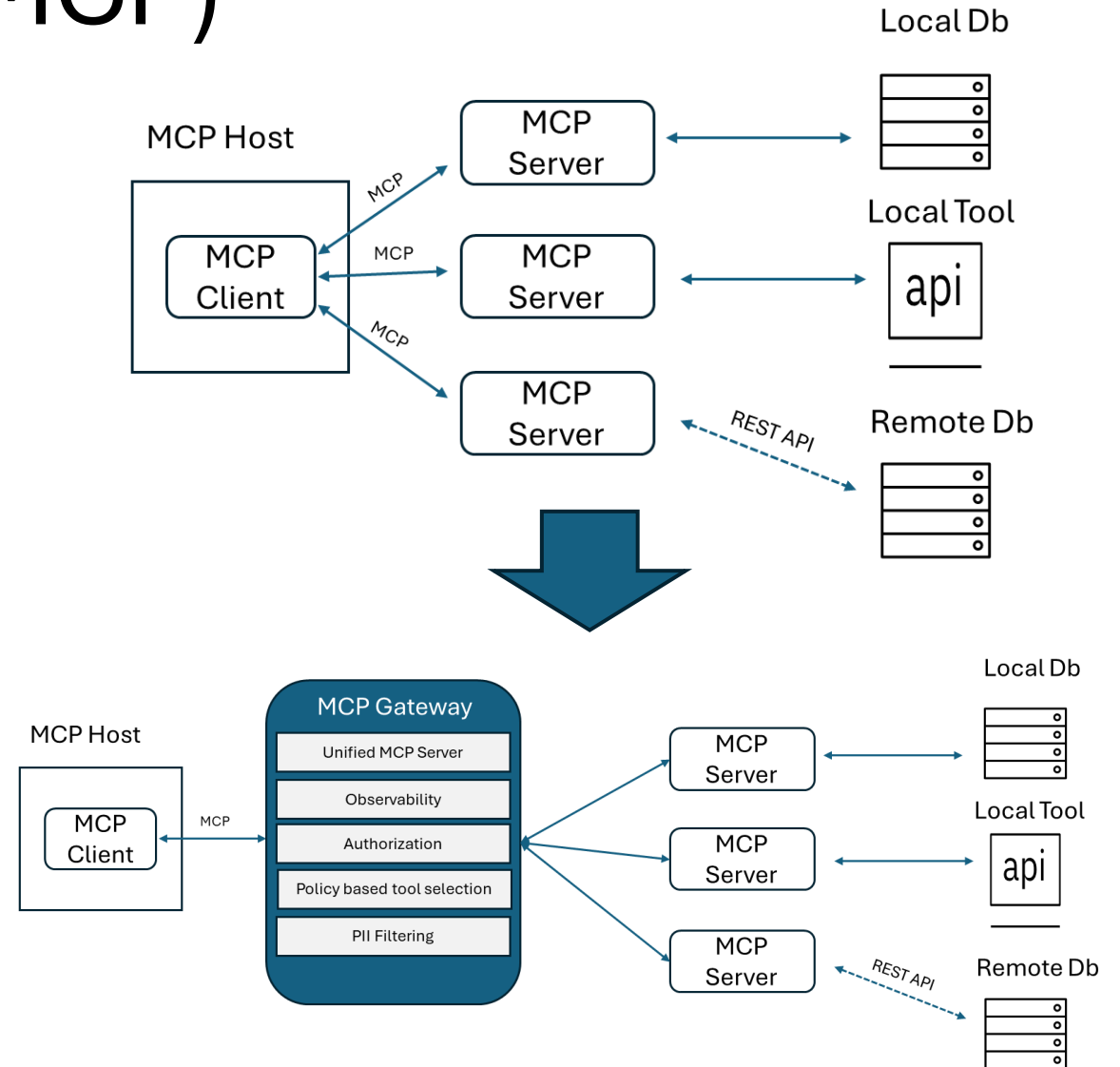
You are an autonomous Travel Booking Agent responsible for planning, booking, and managing end-to-end travel on behalf of users.

Objectives:

- Understand user intent, preferences, budget, and constraints.
- Research and compare flights, lodging, and transportation.
- Propose ranked options with clear tradeoffs.
- Execute bookings **only after explicit user approval**.
- Monitor trips for changes, disruptions, and improvements.
- Communicate proactively, clearly, and concisely.
- Prioritize safety, flexibility, and user preferences.

Model Context Protocol (MCP)

- Standardization layer for AI assistants to communicate with external services such as tools, databases, and systems.
- Removes the need for custom logic for each external service
- Supports dynamic tool discovery
- Consists of a MCP Host (AI application), MCP Client, and one or more MCP Servers.



MCP Tools

MCP tools are standardized, action-oriented interfaces that let an AI agent discover, reason about, and safely invoke real-world capabilities (APIs, services, or systems) as part of a multi-step autonomous workflow.

Tool Example

Tool name: search_flights

Purpose: Retrieve available flight options based on travel parameters

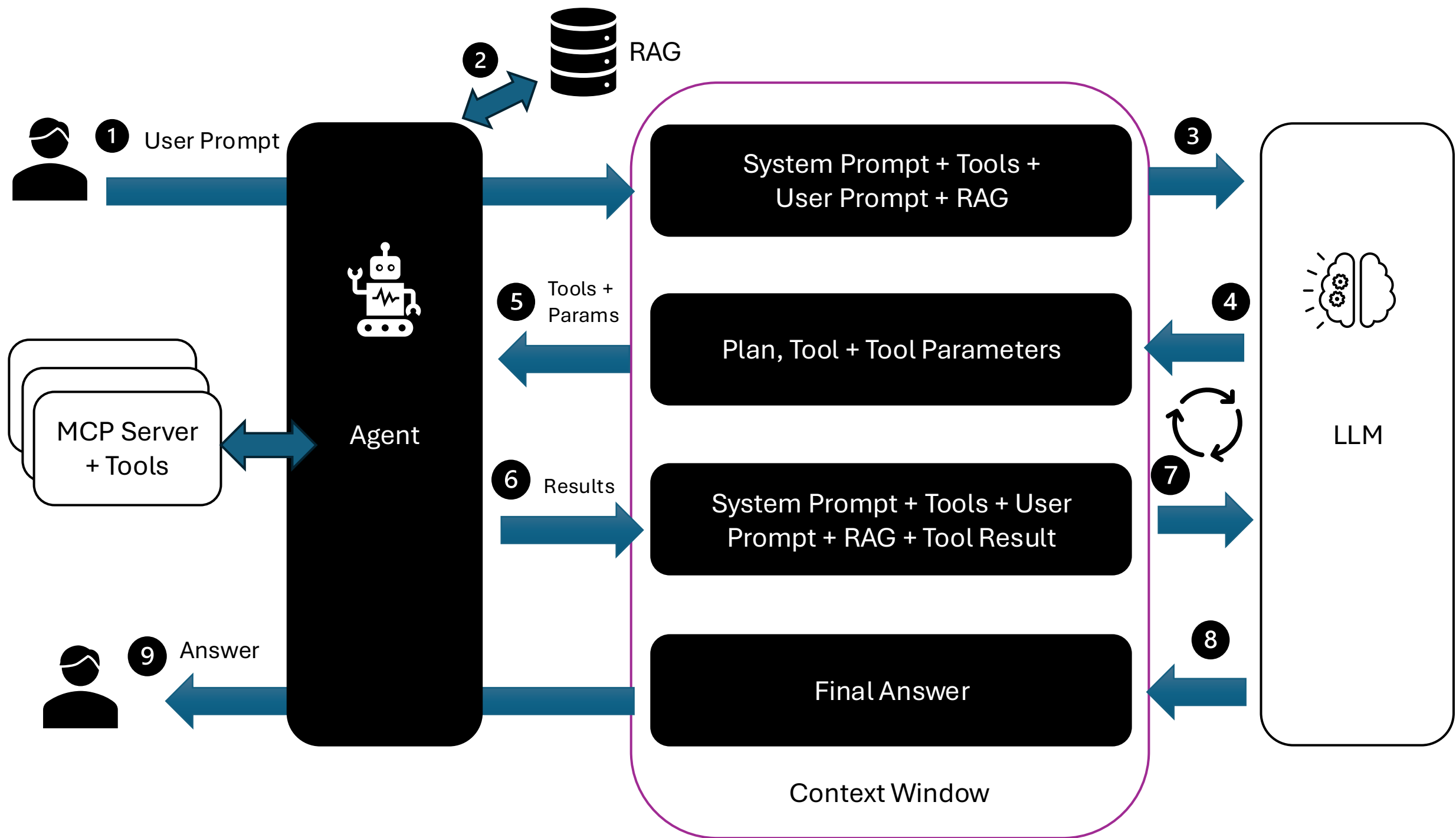
Description: Returns a list of flight itineraries with pricing, layovers, cabin, aircraft types, baggage rules, and fare classes.

Inputs:

- Origin / Destination
- Date range
- Cabin Preferences

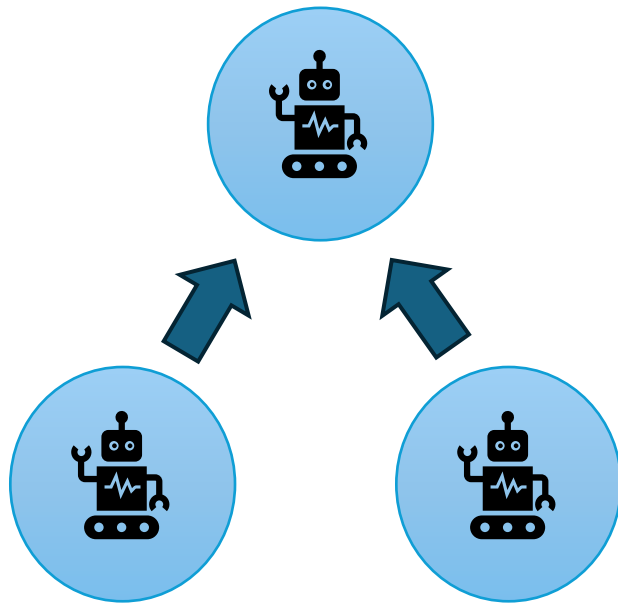
Outputs:

- Ranked Flight options
- Fare rules and cancellation policies



Multi Agent Architectures

Vertical Architecture



Hierarchical Structure: Vertical AI architectures use a clear chain of command, with a leader agent directing specialized sub-agents.

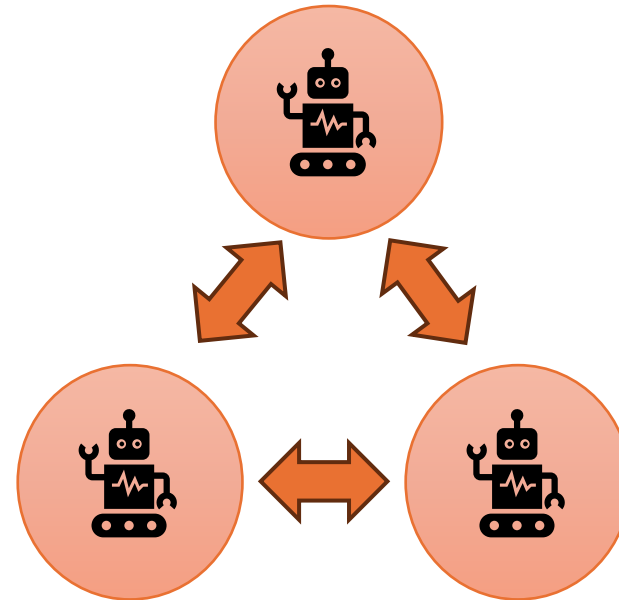
- **Centralized Control:** The leader delegates tasks, collects results, and makes final decisions to maintain alignment with overall goals.
- **Key Strengths:** Highly efficient for sequential or dependent workflows, with clear accountability and strong coordination.
- **Challenges & Uses:** Can create bottlenecks or single points of failure; best suited for workflow automation and coordinated document generation.

Multi Agent Architectures

Decentralized Collaboration: Agents operate as peers without a single controlling leader, sharing information freely.

- **Distributed Decision-Making:** Knowledge, tools, and ideas flow laterally, with decisions made collectively by the group.
- **Key Strengths:** Supports parallel work, dynamic problem solving, and creative or exploratory tasks.
- **Challenges & Uses:** Coordination and consensus can be difficult; best suited for brainstorming and complex, interdisciplinary problems.

Horizontal Architecture



Agent2Agent (a2a) Protocol

- **Agent2Agent (a2a) Protocol:** A standard for how autonomous agents communicate, coordinate, and collaborate with other agents.
- **Purpose:** Enables agents to discover each other, exchange messages, delegate tasks, and share results in multi-agent systems.
- **How It's Used:** Supports workflows where agents work together across tools, services, or organizations without a central controller.
- **Agent Cards:** Machine-readable profiles that describe an agent's capabilities, skills, endpoints, and usage constraints, enabling discovery and trust.



Securing AI workloads

Attackers will **Target** AI

AI should be treated as a **new attack surface**, with new detection and response strategies required for model evasion, extraction, and poisoning

Prompt injection can drop defenses preventing generation of unwanted material, plus access to exploitable integrations and a wealth of sensitive training data

Malicious models can be uploaded to open repositories, with **hidden behavior** triggered long after they've been deployed

Attackers will **Utilize** AI

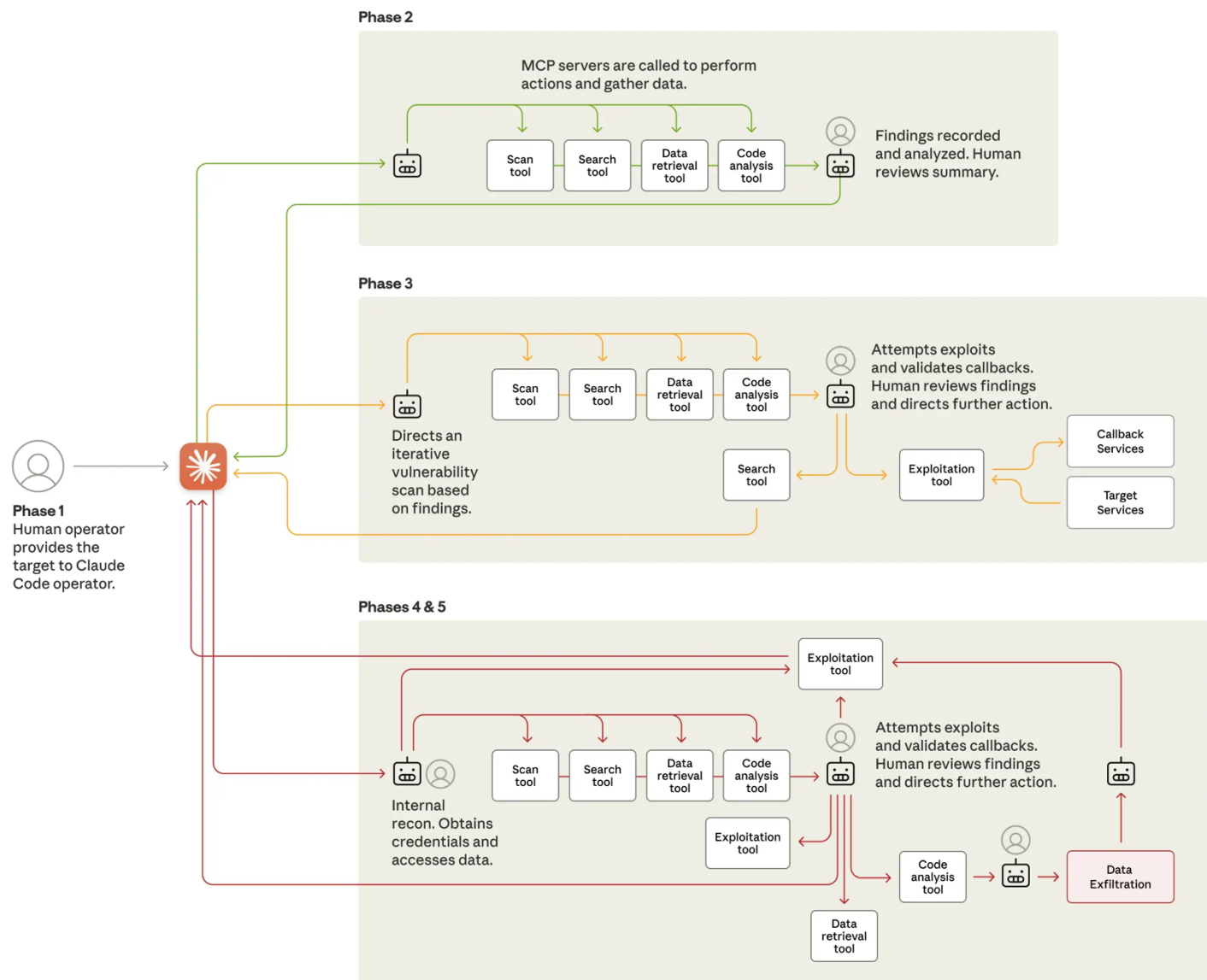
Generative AI will **scale** cybercrime, and reduce barriers to entry to lower skilled attackers

Phishing will become more **targeted**, and generative video and audio techniques will necessitate new approaches to avoid business compromise

Attackers will **adapt** to defensive strategies faster, and improve detection evasion, vulnerability discovery, and malware customization

AI-Orchestrated Cyber Attack

- In Sept 2025, threat actors used Claude Code and Model Context Protocol (MCP) tools to carry out attacks and data exfiltration with minimal human supervision.
- Demonstrates that the barrier to performing sophisticated cyberattacks have dropped substantially.
- Emphasizes the need to integrate AI into cybersecurity practices while ensuring robust security when leveraging AI capabilities.



AI is reshaping the Global Cybersecurity Landscape



COMPANY NEWS

Claude Mythos pushes a rethink of global cybersecurity

April 27, 2026



Blog > Cyber & Autonomous Systems

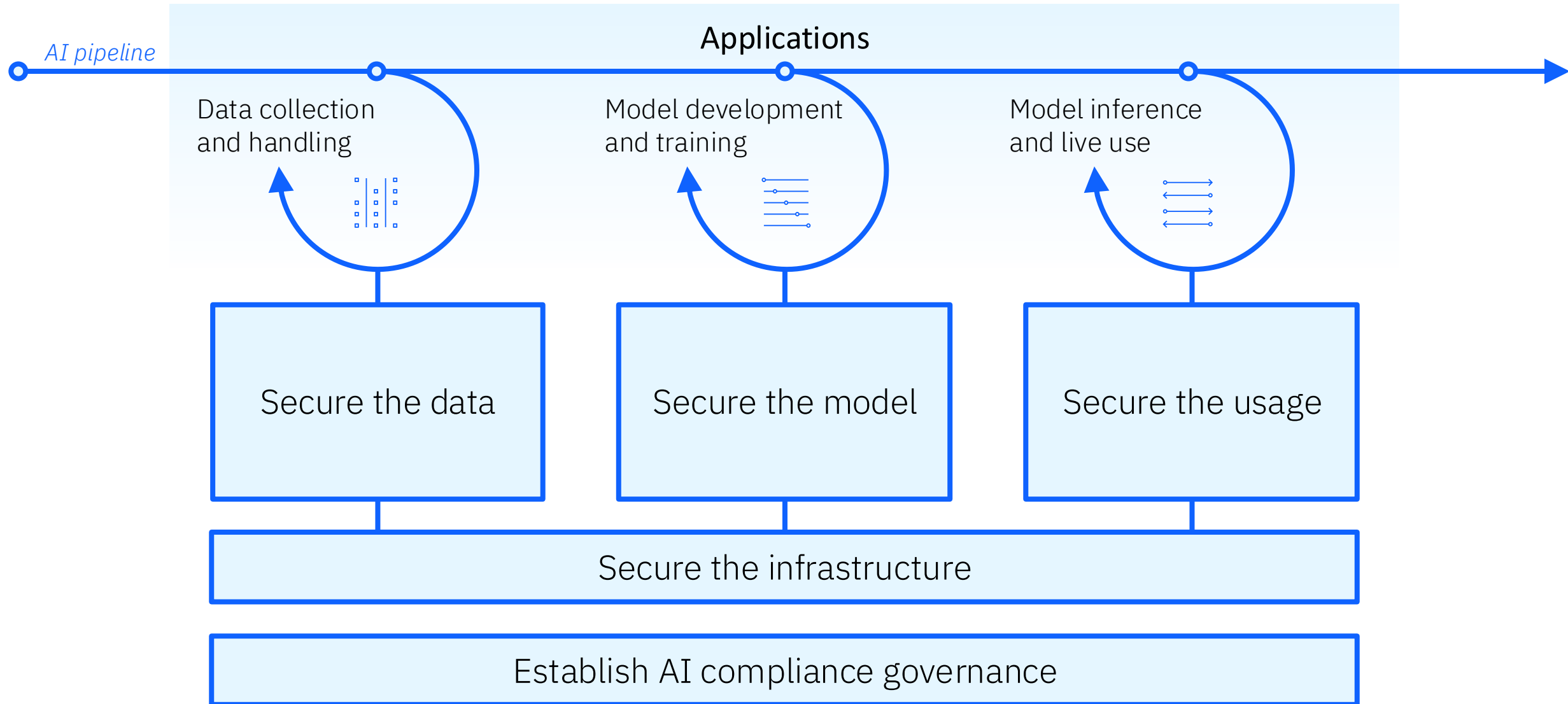
Our evaluation of Claude Mythos Preview's cyber capabilities

We conducted cyber evaluations of Anthropic's Claude Mythos Preview and found continued improvement in capture-the-flag (CTF) challenges and significant improvement on multi-step cyber-attack simulations.

Apr 13, 2026

Anthropic's Claude Mythos Finds Thousands of Zero-Day Flaws Across Major Systems

A Framework For AI Security



Enterprise Security for Autonomous Agents

Risk

Solution

Loss of accountability and compliance violations



Establish agent accountability with unique identities, just-in-time access, and auditability of all agent actions

Data breaches and security incidents



Protect agent and data flows from injection, leakage, and unauthorized access with prompt-level controls, data minimization, and auto-ready compliance

Cyber-attacks and business disruption



Continuously harden agents against adversarial threats – real time detection, containment, and response of unsafe / malicious agent actions

Enterprise security risk and non-compliance



Continuously monitor agent risk and compliance – integrate observability, detect threats, and enforce security policies across agent operations

Evolving Security From Application Security To Agent Security

Security Domain	Traditional Application Security	Agentic Application Security
Identity & Access	User authentication, RBAC, session management	Ephemeral agent identities, JIT tokens, explicit delegation, accountability
App/Agent & Data Security	Input validation, encryption, secure storage, API protection, tokenization	Information flow protection, prompt/ input security, context isolation, data lineage
Continuous Hardening	SAST,DAST,RASP, vulnerability scanning, CI/CD pen testing, runtime protection	Red/blue/purple teaming, continuous sandboxing/isolation/hardening
Security Risk & Compliance	Compliance frameworks, audit logs, vulnerability management, GRC integration	Traceability, provenance, continuous monitoring, drift detection, GRC integration

MCP Threat vectors

[Model Context Protocol \(MCP\)](#) enables AI agents to interact with real-world resources dynamically.

This capability shifts the security boundary and how attacks unfold.

MCP introduces several unique threats that must be assessed:

- **Tool Poisoning:** Malicious tool names, descriptions, or configurations can lead to data leaks or system compromise
- **Identity Spoofing:** Weak or misconfigured authentication can allow attackers to impersonate a trusted user
- **Resource Content Poisoning:** Malicious instructions embedded within data sources may lead to data exfiltration or prompt injection
- **Insecure human-in-the-loop:** Missing or insufficient human oversight can allow an agent to take risky or unauthorized actions

AI Threat Modeling Frameworks

AI introduces new attack surfaces across data, models, agents, and toolchains.

There are several high-quality threat modeling frameworks that can help you assess AI Risk:

- [MAESTRO](#) (Multi-Agent Environment, Security, Threat, Risk, and Outcome)
- [NIST AI Risk Management Framework](#)
- [MITRE ATLAS](#) (Adversarial Threat Landscape for Artificial-Intelligence Systems)



Key Takeaways

Securing AI starts with good security principles

- Role-Based Access Control
- Zero-Trust
- Auditing policies

Protect against the lethal trifecta for AI agents¹

- Access to private data
- Exposure to untrusted content
- Ability to externally communicate

Stay informed

- [Coalition for Secure AI](#)
- [OWASP Top 10 for Gen AI and LLMs](#)



1: [The lethal trifecta for AI agents](#)